

Coordinate Taxi Routes

Unit 9: Treasure Map Navigator · Lesson 9-6 · EduWonderLab

UNIT 9 · LESSON 9-6 · TEACHER NOTES

Standard 6.NS.8

FORMAT Route distance task cards

TIME 25–30 min

GROUPING Pairs

TOPIC Distance on the Coordinate Plane

STANDARD 6.NS.8

PREP LEVEL Light — print route maps and point cards

Learning goal *I can find the distance between two points that share an x- or y-coordinate.*

Teacher setup

- Print one set per pair. Each card gives two points on the same horizontal or vertical line.
- Students find the distance using absolute value.
- Compare the straight grid distance to the route.

Materials

Route maps, Coordinate-point cards, Absolute value distance organizer, Pencil.

Differentiation & language support

- Reminder: distance = |difference of the changing coordinate| (SPED).
- Sentence stem: “The distance is ___ because $|__ - __| = __$.”
- ESOL: count the units along the grid line.

Key vocabulary (English / Español):

Term	Español	What it means
distance	distancia	how far apart two points are
absolute value	valor absoluto	distance from zero, always positive

Quick teacher check: *Check Card 3 — subtracting a negative coordinate.*

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Name: _____ Date: _____ Class: _____

Your goal: *I can find the distance between two points that share an x- or y-coordinate.*

What you need

Route maps, Coordinate-point cards.

How to play

1. Read the two points on a card.
2. Find which coordinate changes (x or y).
3. Take the absolute value of the difference.
4. Record the distance in units.

Route Cards

For points on the same line, the distance is the absolute value of the difference of the changing coordinate.

1. (2, 3) to (2, 8)	2. (1, 4) to (6, 4)
3. (-3, 2) to (-3, -4)	4. (-5, 1) to (4, 1)
5. (0, 0) to (0, 7)	6. (3, -2) to (3, 5)
7. (-6, -1) to (2, -1)	8. (4, 4) to (4, -4)

Record your work (difference + distance):

Explain your thinking

Explain why you use absolute value to find distance.

Sentence frame: *The distance is ___ because $|___ - ___| = ___$.*

★ **Challenge:** Two points are $(-2, 5)$ and $(-2, -3)$. Find the distance, then describe the route. (8 units)

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. (2, 3) to (2, 8) Answer: 5 units ($ 8 - 3 $)	2. (1, 4) to (6, 4) Answer: 5 units ($ 6 - 1 $)
3. (-3, 2) to (-3, -4) Answer: 6 units ($ 2 - (-4) $)	4. (-5, 1) to (4, 1) Answer: 9 units ($ 4 - (-5) $)
5. (0, 0) to (0, 7) Answer: 7 units	6. (3, -2) to (3, 5) Answer: 7 units ($ 5 - (-2) $)
7. (-6, -1) to (2, -1) Answer: 8 units ($ 2 - (-6) $)	8. (4, 4) to (4, -4) Answer: 8 units ($ 4 - (-4) $)

Teaching notes & look-fors

- Same $x \rightarrow$ vertical distance; same $y \rightarrow$ horizontal distance.
- Card 3: $|2 - (-4)| = |6| = 6$ units.